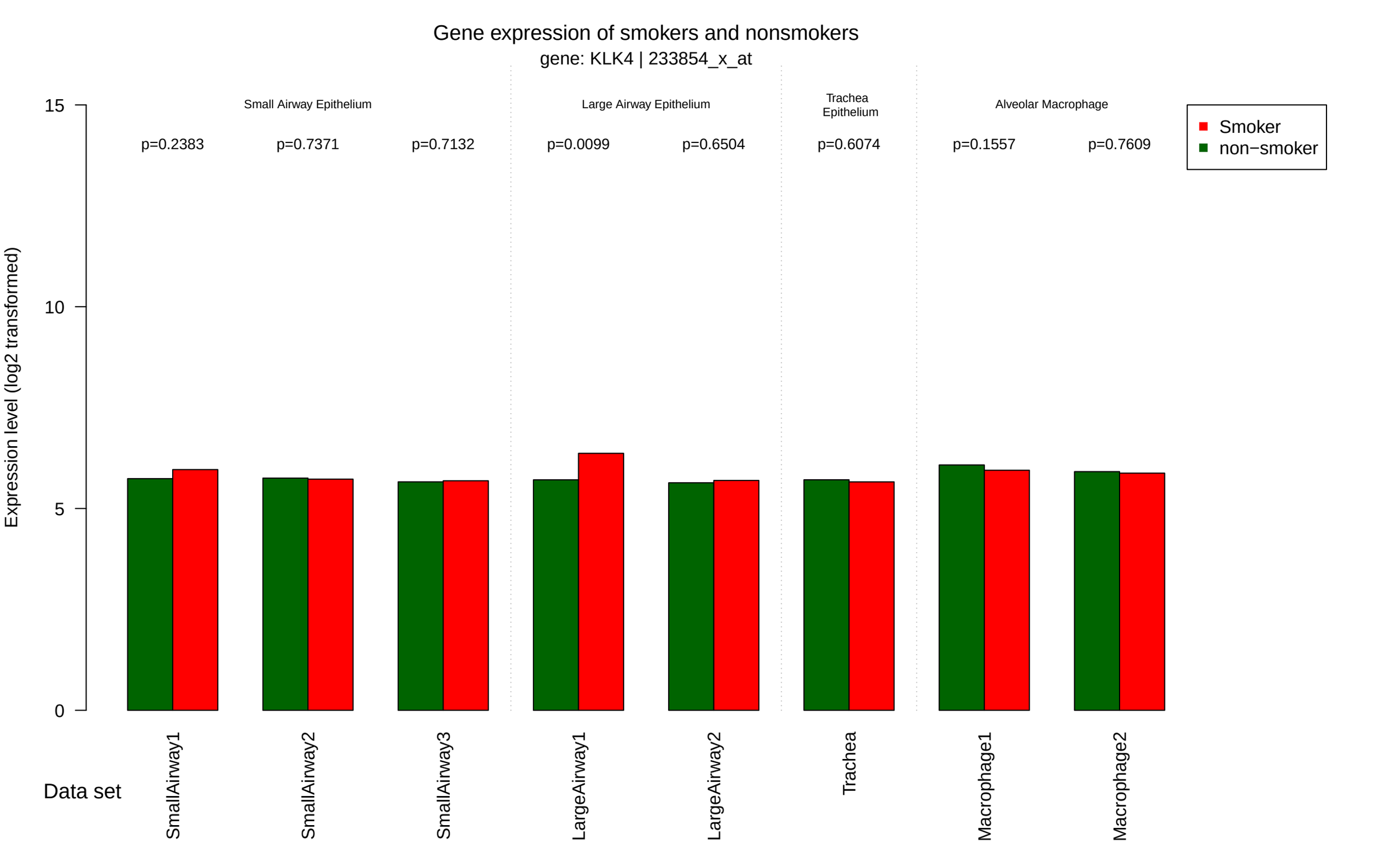
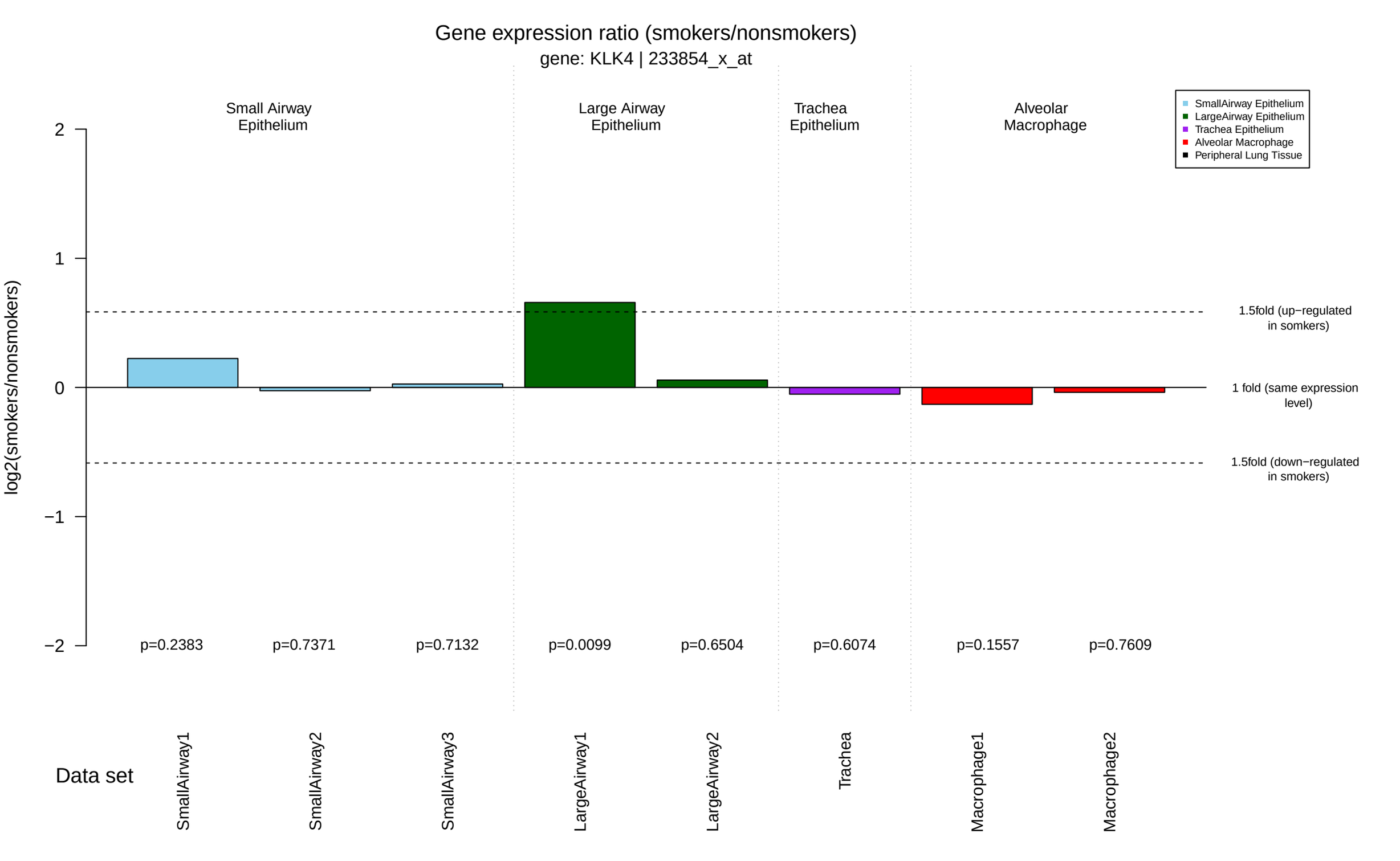
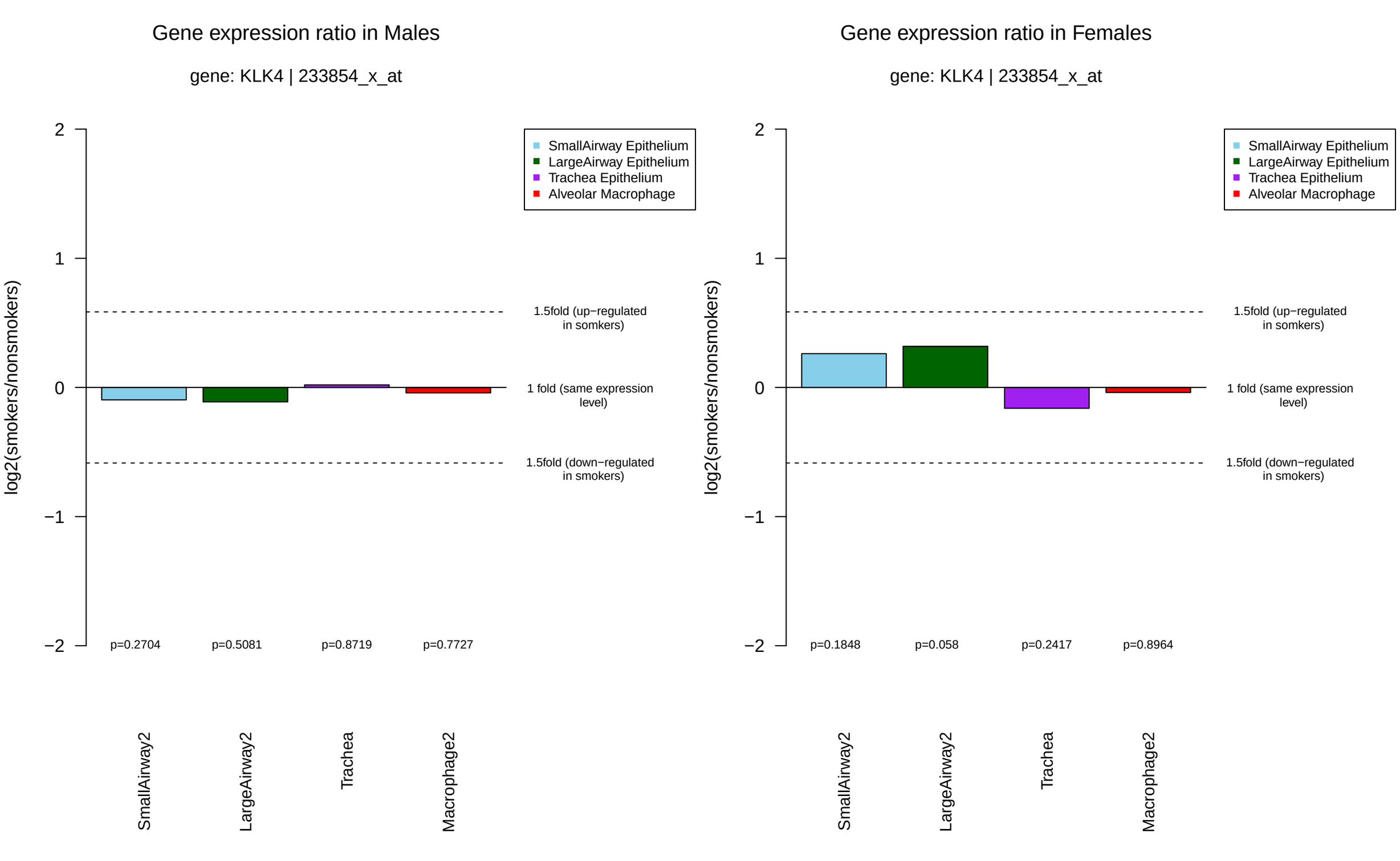
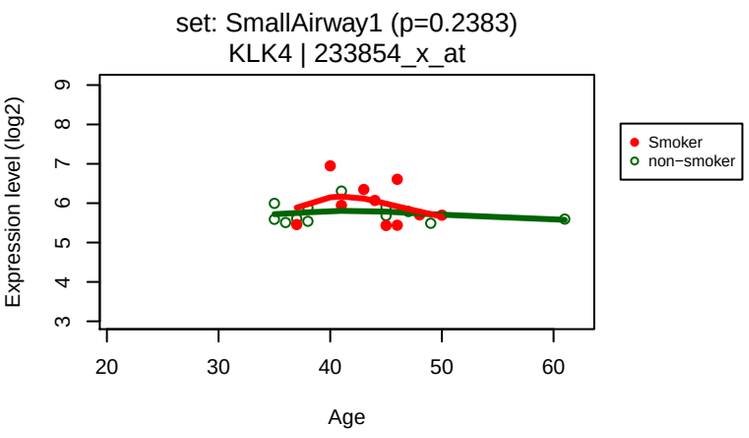
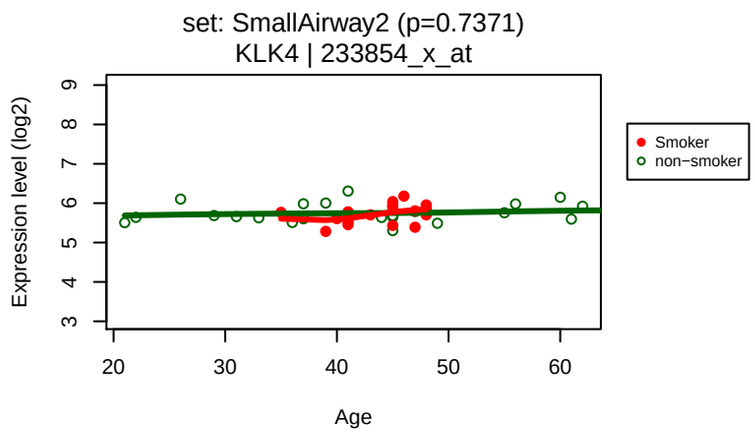
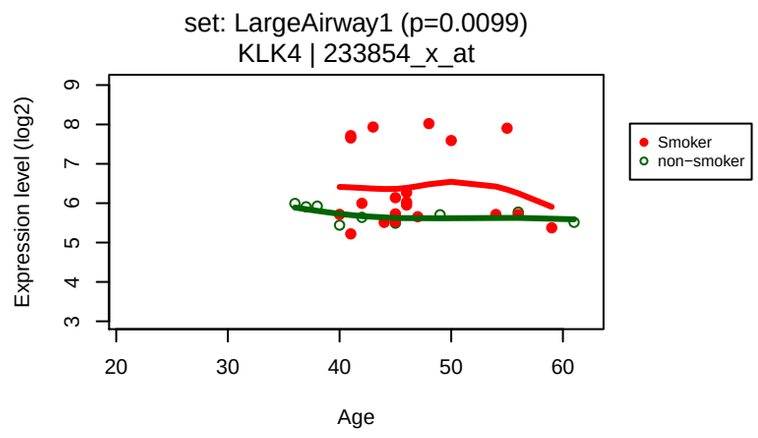

Gene expression of smokers and nonsmokers
gene: KLK4 | 233854_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2383
p=0.7371
p=0.7132
p=0.0099
p=0.6504
p=0.6074
p=0.1557
p=0.7609
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

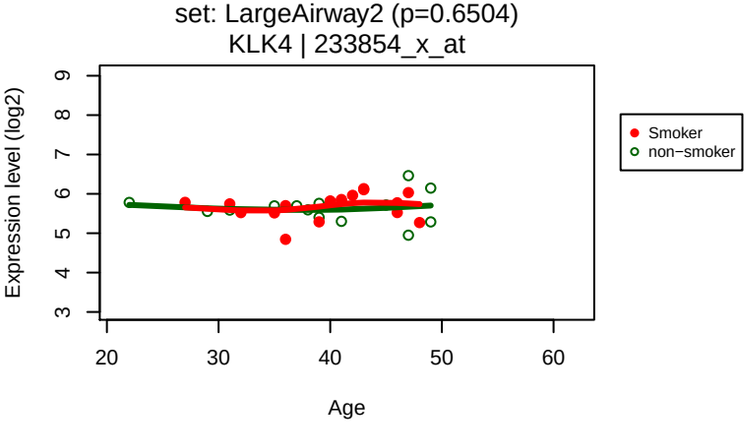

Gene expression ratio (smokers/nonsmokers)
gene: KLK4 | 233854_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2383
p=0.7371
p=0.7132
p=0.0099
p=0.6504
p=0.6074
p=0.1557
p=0.7609
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

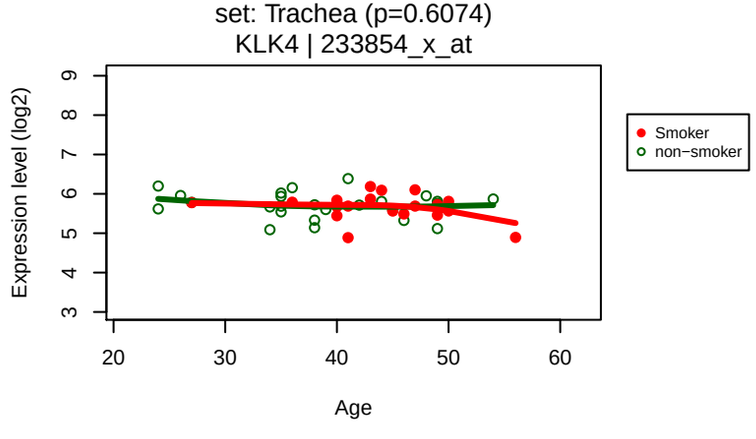

Gene expression ratio in Males
gene: KLK4 | 233854_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2704
p=0.5081
p=0.8719
p=0.7727
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KLK4 | 233854_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1848
p=0.058
p=0.2417
p=0.8964
SmallAirway2
LargeAirway2
Trachea
Macrophage2

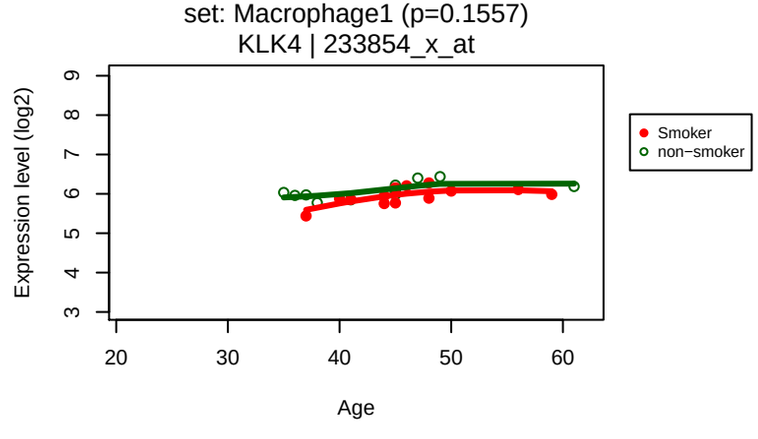

set: SmallAirway1 (p=0.2383)
KLK4 | 233854_x_at
Expression level (log2)
Age
Smoker
non−smoker

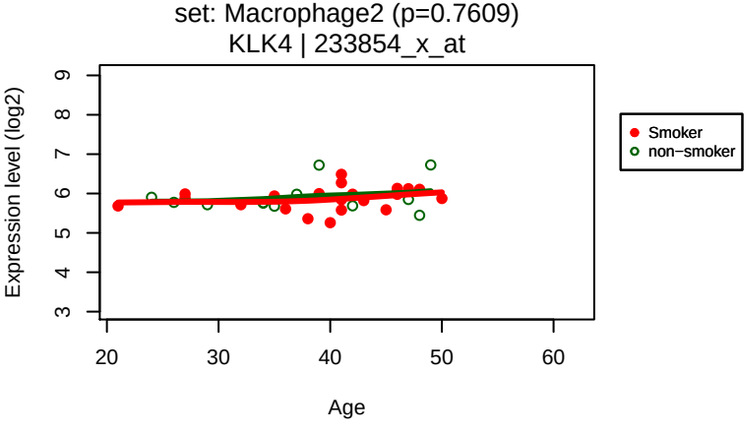

set: SmallAirway2 (p=0.7371)
KLK4 | 233854_x_at
Expression level (log2)
Age
Smoker
non−smoker

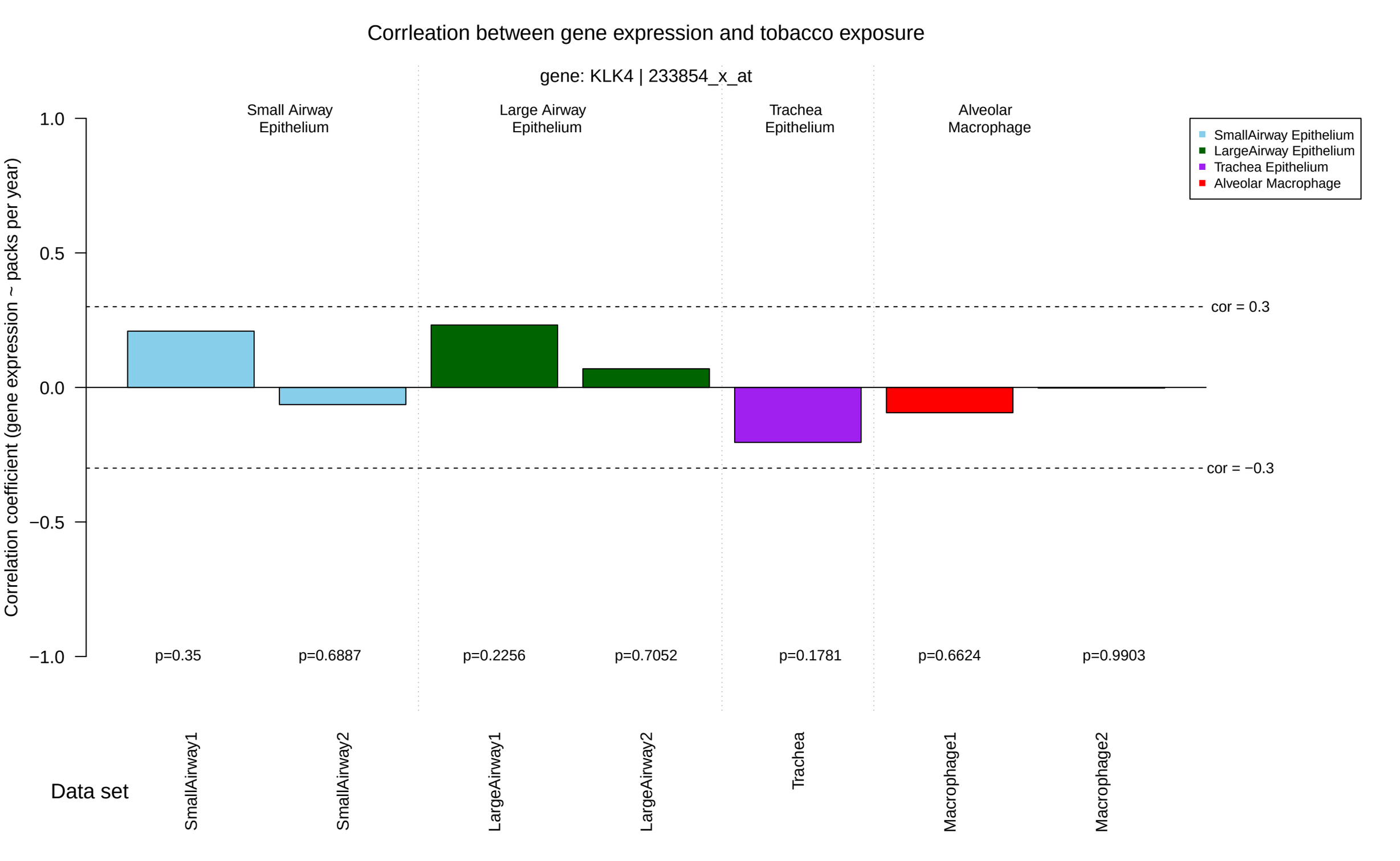

set: LargeAirway1 (p=0.0099)
KLK4 | 233854_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6504)
KLK4 | 233854_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6074)
KLK4 | 233854_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1557)
KLK4 | 233854_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7609)
KLK4 | 233854_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KLK4 | 233854_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.35
p=0.6887
p=0.2256
p=0.7052
p=0.1781
p=0.6624
p=0.9903
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2